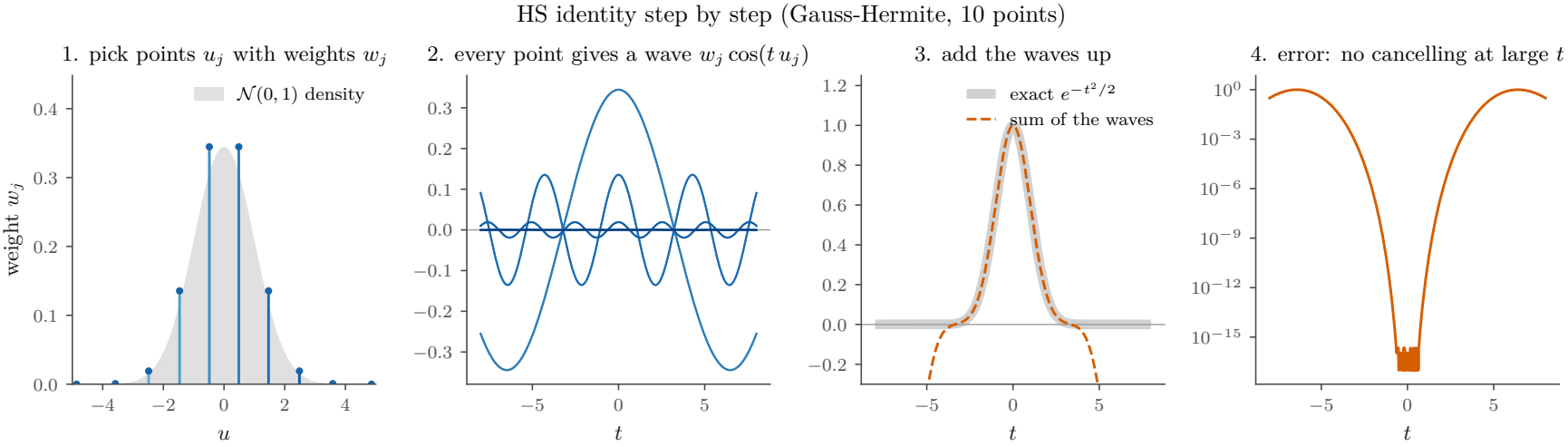

HS identity step by step (Gauss-Hermite, 10 points)
1. pick points u_j with weights w_j
$\mathcal{N}(0,1)$ density
weight w_j
u
2. every point gives a wave $w_j \cos(t\,u_j)$
t
3. add the waves up
exact $e^{-t^2/2}$
sum of the waves
t
4. error: no cancelling at large t
t